Hao Jiang

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Education

University of Electronic Science and Technology of China (UESTC) **2024.9 - 2028.6**
Yingcai Honors College **Major:** Fundamental Science of Mathematics & Physics (Elite Talent Class) **Focus:** AI
GPA: 3.96/4.0 **Weighted Average: 91.55** **Rank: 2/24** **CET-4: 581** **CET-6: 561**
Core Courses: Advanced Calculus, Advanced Algebra and Geometry, University Physics, Probability and Mathematical Statistics, Data Structures, Advanced Programming Language Design, Machine Learning, etc.

Research Experience

➤ **MLLM reasoning and post-training** | Thinking with latent visual memory **2026.4 - 2026.10**

ReMAP: Restoring the Perceptual Cycle with Reasoning-Time Latent Visual Memory [arXiv]
Advisor: Asst. Prof. Shanghang Zhang, School of Computer Science, PKU Submitted to ICLR 2027 **First Author**

- **Finding:** Grounding fades as MLLMs reason longer. Analyzing visual memory (curation, organization, access): local evidence needs global context, compact latents balance accuracy and cost, access utility depends on reasoning state.
- **Method & results:** ReMAP couples a question-conditioned Global memory and a state-driven Local memory (both compact latent tokens) with an RL policy deciding when to invoke which. Beats prior visual-memory methods on all four multi-image benchmarks (+8.38/+14.84 on MuirBench/MIMIC) and cuts visual tokens by 51.0–76.8%.

➤ **MLLM safety** **2025.9 - 2025.12**

Persuasion in Scene: A Multi-Agent Typographic Jailbreak Crew against Large Vision-Language Models [pdf]
Submitted to CVPR 2026 **Co-First Author**

Models as Lego Builders: Assembling Malice from Benign Blocks via Semantic Blueprints [pdf]
Accepted by CVPR 2026 **Contributor**

- PIS builds scenario-grounded typographic prompts with cooperating agents; StructAttack shows that once a harmful query is split into locally benign semantic slots, the model's own compositional reasoning reassembles the malicious intent, enabling a single-query, optimization-free black-box attack. Both lead in attack success rate.
- **Contribution:** PIS co-first author (ideation through writing); StructAttack contributor (SafeBench evaluation).

➤ **AI for pathology** | Multiple instance learning **2025.3 - 2025.8**

CKMIL: Cascaded Key-Instance Attention Multiple Instance Learning for Histopathology Whole Slide Image Analysis [pdf]
Submitted to AAAI 2026 **First Author**

Advisors: Prof. Liang-Jian Deng, School of Mathematical Sciences, UESTC Asst. Prof. Hao Chen, HKUST

- **Problem & method:** Diagnostic evidence in a whole slide image (WSI) sits in a few patches, and global interaction dilutes it. Proposed CKMIL, a cascaded key-instance attention framework in which key instances guide efficient dependency modeling; leading results on multi-cohort cancer subtyping and survival prediction (BRACS, TCGA).
- **Contribution:** Project lead; drove data processing, method design, experiments, and paper writing.

➤ **AI for medicine** | Pathology image compression **2025.12 - Present**

From Full-Bag Diagnosis to Budgeted Patch Selection: Role-Decoupled Benchmarking and Adaptation of Pathology Foundation Models in Whole-Slide Imaging
Target: IEEE TMI 2027 **First Author**

Advisors: Prof. Liang-Jian Deng, School of Mathematical Sciences, UESTC Asst. Prof. Hao Chen, HKUST

- **Finding & method:** How much of a gigapixel slide does diagnosis need? From pathology image compression, found some morphological tasks need no image reconstruction, owing to image redundancy and foundation model (FM) robustness. Split FMs into Selector and Consumer roles, built a role-decoupled benchmark, and proposed adaptations.

Current direction: After the latent visual memory project, joined the memory world model and spatial memory projects to learn world models and AI infrastructure, while continuing to reflect on "compression is intelligence."

Skills

- **Tools:** Python, PyTorch, Linux, Git, LaTeX, Codex, Claude Code; idea validation, experiments, writing.
- **Deep Learning:** Transformer, Mamba; test-time training, post-training, and evaluation of multimodal LLMs.

Honors & Awards

- UESTC Outstanding Student Scholarship First Class **2025**
- UESTC Outstanding Student Scholarship First Class **2026**
- Mathematical Contest in Modeling (MCM/ICM) Meritorious Winner **2026**
- UESTC (iFLYTEK Cup) 1st AI Agent Creativity Competition First Prize **2026**